SNEH DESAI

Data Scientist | Machine Learning | Data Analytics

sneh14022002@gmail.com | +1 864-356-8926 | Simpsonville, SC
linkedin.com/in/sneh-desai-370987249 | github.com/Sneh14202 | sneh14202.github.io
Date of Birth: 14th February 2002

PROFESSIONAL SUMMARY

Data scientist and M.S. Informatics graduate student with a B.Tech in Computer Science (AI/ML specialization). Builds end-to-end machine learning solutions in Python – from EDA, feature engineering and model tuning to calibration, SHAP explainability and Streamlit deployment – and communicates insights through Power BI dashboards. Currently working in institutional research and analytics at the University of South Carolina.

EXPERIENCE

Math Tutor 08/2026 – Present
University of South Carolina, Math Lab
- Tutor university students in mathematics in the Math Lab, explaining concepts and guiding problem-solving.

Data Analyst Intern 03/2026 – Present
University of South Carolina, Institutional Planning and Research
- Support institutional data analysis and reporting, applying analytical and visualization techniques to identify trends and communicate findings clearly.
- Organize, interpret, and present data to support institutional planning and research.

Graduate Assistant 01/2026 – Present
University of South Carolina, Student Success Center
- Support student-focused programs, coordination, and administrative tasks; collaborate with staff and campus stakeholders while maintaining organized records.

Summer Intern, Data Analytics 07/2023 – 08/2023
Edunet Foundation
- Worked on real-world projects in Power BI and SAP Analytics Cloud, with exposure to machine learning, big data, and cloud computing.

EDUCATION

M.S. in Informatics 2025 – Present
University of South Carolina Upstate

B.E./B.Tech. in Computer Science & Engineering (AI/ML) 2020 – 2024
RNG Patel Institute of Technology

TECHNICAL SKILLS

Machine Learning: Scikit-learn, Gradient Boosting, Hyperparameter Tuning, Probability Calibration, Threshold Optimization, Class Imbalance Handling, SHAP
Programming & Tools: Python, SQL, Pandas, NumPy, Streamlit, pytest, Jupyter Notebook, Microsoft Excel
Analytics & BI: Exploratory Data Analysis, Feature Engineering, Data Visualization, Power BI, SAP Analytics Cloud
Additional: Big Data, Cloud Computing

PROJECTS

Customer Churn Prediction & Retention Intelligence Platform
Python, Scikit-learn, Pandas, SHAP, Streamlit
- Built an end-to-end churn prediction platform on 7,043 telecom customer records, covering EDA, feature engineering, model comparison, hyperparameter tuning, probability calibration, and threshold optimization.
- Validated a calibrated Gradient Boosting model with 0.846 ROC-AUC, 0.674 PR-AUC, 81.3% recall, and 0.626 F1; risk segmentation flagged 28.3% of customers as high/critical risk.
- Applied SHAP explainability and business-value thresholding to generate customer-level retention recommendations; deployed an interactive Streamlit risk-scoring application.

Fraud Detection & Transaction Risk Intelligence Platform
Python, Scikit-learn, SHAP, Streamlit
- Built an end-to-end fraud detection pipeline on 284,807 credit card transactions, addressing severe class imbalance via model comparison, hyperparameter tuning, threshold optimization, and probability calibration.
- Engineered transaction and time-based features and developed reusable Python modules for feature engineering, risk scoring, and model inference, backed by automated pytest tests.
- Implemented SHAP-based transaction explanations, five-level fraud risk segmentation, investigation prioritization, and hypothetical business-value analysis; deployed as an interactive Streamlit app.

Power BI Dashboards
Game Sales, Adidas Sales, Swiggy Orders
- Built dashboards analyzing video game sales by genre and year, Adidas sales by product type and season, and Swiggy order volume by city and cuisine using historical data.

CERTIFICATIONS & LANGUAGES

Certifications: IBM SkillsBuild – Data Analytics (2-week virtual internship); Data Science Tools – Jupyter Notebook, Power BI, Microsoft Excel
Languages: English, Hindi, Gujarati